OpenClaw Peripheral Hardware Configuration

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1. Course Overview

This course will introduce OpenClaw's default hardware configuration files before use, focusing on how to modify the underlying default parameters based on the actual peripherals you have, and completing basic verification.

- Use underlying commands to verify whether modifications take effect

2. Hardware Default Configuration File Description

2.1 Configuration File Location

OpenClaw's current hardware default configuration file is located at:

```
/home/jetson/hardware_hal_demo/hardware_config.py
```

If you are just following the accompanying tutorials and using the factory hardware, you usually don't need to modify it.

2.2 Current Configuration File Coverage

`hardware_config.py` currently mainly contains three types of default configurations:

- OpenClaw firmware serial port default parameters
- OLED default parameters
- CSI camera or USB camera

These default values will be read by underlying command-line tools, for example:

- `rgb`
- `oled`
- `openclaw_adapter`
- `CAMERA_DEFAULTS`

Note:

- **DHT and RGB are currently driven directly through OpenClaw firmware**
- **OLED is driven through the mainboard I2C pins**
- **CSI camera/USB camera needs to check hardware connection**

3. DHT Current Configuration Features

3.1 Verification Method

```
dht   #Get sensor data
dht_oled  #Display sensor data to OLED
```

Normal terminal input, you can see OLED displaying temperature and humidity data



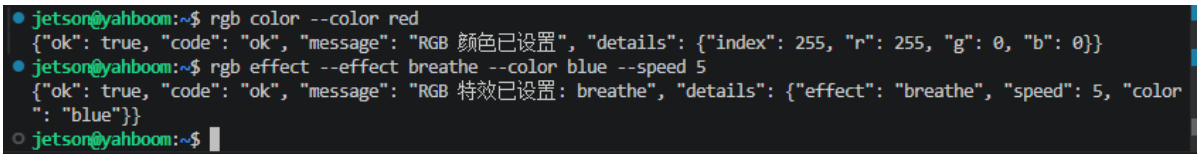
4. RGB Default Configuration

4.1 Verification Method

After saving, run in terminal:

```
rgb color --color red #All turn red
rgb effect --effect breathe --color blue --speed 5 #Blue breathing light
```

Normal terminal input, you can see RGB lights displaying normally



5. OpenClaw Firmware Serial Port Default Configuration (Servo Verification)

5.1 Current Default Values

Current OpenClaw firmware serial port default parameters are:

- `port="/dev/ttyTHS1"`
- `baudrate=115200`

5.2 Modification Example

Edit configuration file:

```
home/jetson/hardware_hal_demo/hardware_config.py
```

Change the values for your board, for example:

```
"port": "/dev/ttyTHS1",
"baudrate": 115200,
```

5.3 Verification Method

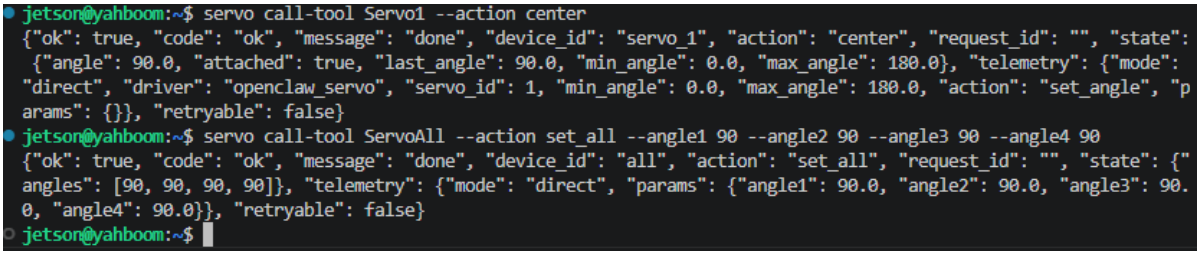
After saving, it is recommended to first use servo or RGB underlying commands for testing, for example:

```
rgb color --color green
```

Or:

```
servo call-tool Servo1 --action center  #Servo 1 center to 90 degrees
# All servos return to center position
servo call-tool ServoAll --action set_all --angle1 90 --angle2 90 --angle3 90 --
angle4 90
```

If the command can return normally and control peripherals normally, it indicates the serial port link is available.



6. OLED Default Configuration

6.1 Current Default Values

Current OLED default parameters mainly include:

- `width=128`
- `height=32`
- `interface="i2c"`
- `i2c_bus="auto"`
- `i2c_bus_candidates=[1, 0, 7, 8]`
- `i2c_address=0x3C`
- `rotate=0`
- `invert=False`
- `contrast=255`

6.2 Scenarios Suitable for Modification

It is recommended to modify OLED default values in the following situations:

- You are using an OLED with different resolution
- I2C address is not `0x3C`
- Screen orientation needs to be flipped

6.3 Common Modification Items

Modify Resolution

For example, change:

```
"width": 128,
"height": 32,
```

To:

```
"width": 128,
"height": 64,
```

Modify I2C Address

For example, change:

```
"i2c_address": 0x3C,
```

To:

```
"i2c_address": 0x3D,
```

Modify Rotation Direction

For example, change:

```
"rotate": 0,
```

To:

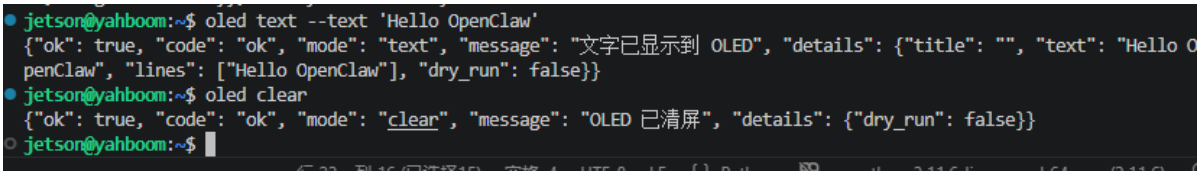
```
"rotate": 180,
```

6.4 Verification Method

After saving, run in terminal:

```
oled text --text 'Hello OpenClaw'
oled clear
```

If OLED can display and clear normally, it indicates configuration has taken effect.



7. Camera Hardware Configuration

7.1 Configuration

Prerequisite is to ensure CSI camera is connected. The current CSI camera pipeline mainly works through the following two entry points:

If using USB camera, you can modify the configuration of this file:

```
/home/jetson/hardware_hal_demo/hardware_config.py
```

```
# Camera default configuration
# backend: "csi" uses Jetson CSI camera; "usb" uses USB camera
CAMERA_DEFAULTS = {
    "backend": "usb",
    "usb_device": "/dev/video0",
    "usb_format": "auto",
    "width": 1280,
    "height": 720,
    "framerate": 30,
    "flip_method": 2,
    "sensor_id": 0,
}
```

Camera CLI command path,

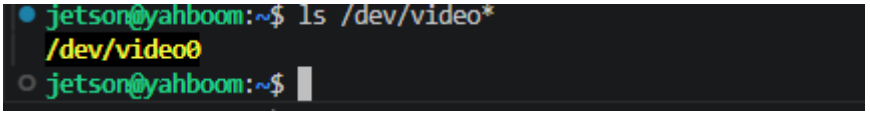
```
/home/jetson/hardware_hal_demo/cli/seewhat
```

7.2 Check Port

Terminal input,

```
ls /dev/video*
```

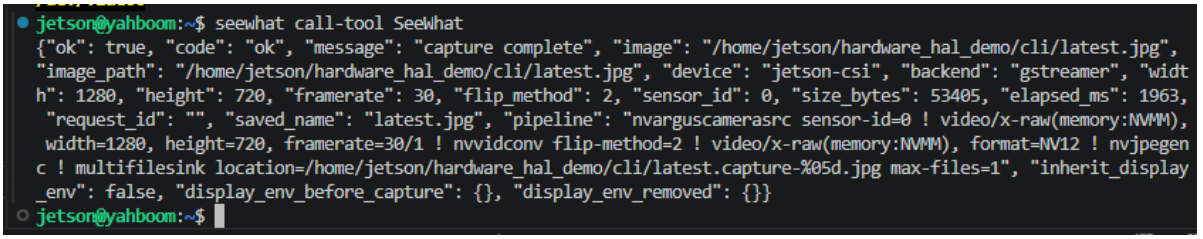
Normally, CSI camera can see one device video0, USB camera can see video0 and video1 devices,



7.3 Photo Verification Method

Terminal input command, first directly test photo capture pipeline:

```
seewhat call-tool SeeWhat
```



If photo capture is successful, you will usually see output files in the following directory:

```
/home/jetson/hardware_hal_demo/cli/latest.jpg
```

